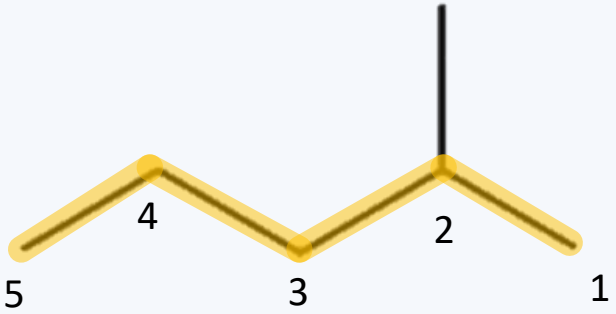
A01 FOUNDATION

C_6H_{14}

1 · NUMBERED PARENT



Parent (5 C) highlighted; carbons numbered from the right-hand end

2 · NAMING TRAIL

Parent: *longest chain = 5 C → pentane*

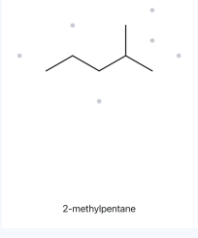
Numbering: *L→R gives 4; R→L gives 2. First point of difference: 2 < 4, so the number from the right*

Substituents: *(A→Z): methyl on C2*

3 · MY NAME (WRITTEN FIRST)

2-methylpentane

4 · THE ALKANE CHECK



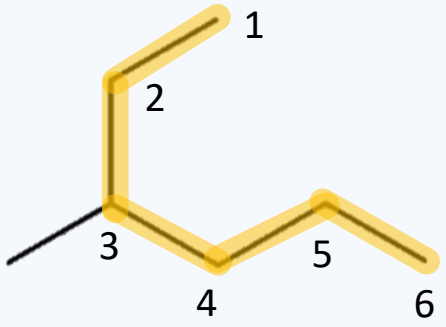
Returned Name: *2-methylpentane*
Verdict: ✓ ***Match***

☑ longest chain/ring ☑ lowest locants ☑ alphabetical order ☑ punctuation ☑ formula counted

A02 FOUNDATION

C_7H_{16}

1 · NUMBERED PARENT



Parent (6 C) highlighted; carbons numbered from the top to right end

2 · NAMING TRAIL

Parent: *longest chain = 6 C → hexane*

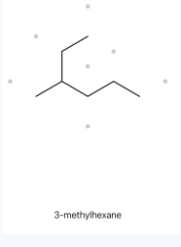
Numbering: *T→ Right End gives 3; Right End → T gives 4. First point of difference: 3 < 4, so the number from the top*

Substituents: *(A→Z): methyl on C3*

3 · MY NAME (WRITTEN FIRST)

3-methylhexane

4 · THE ALKANE CHECK



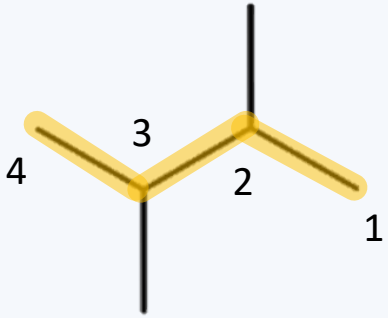
Returned Name: *3-methylhexane*
Verdict: ✓ ***Match***

☑ longest chain/ring ☑ lowest locants ☑ alphabetical order ☑ punctuation ☑ formula counted

A03 FOUNDATION

C_6H_{14}

1 · NUMBERED PARENT



Parent (4 C) highlighted; carbons numbered from the right to left end

2 · NAMING TRAIL

Parent: *longest chain = 4 C → butane*

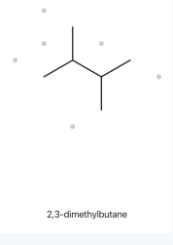
Numbering: *L→R gives 2,3; R→L gives 2,3. Both directions give the same set, so either works. Number from right to left*

Substituents: *(A→Z): methyl on C2; methyl on C3*

3 · MY NAME (WRITTEN FIRST)

2,3-dimethylbutane

4 · THE ALKANE CHECK



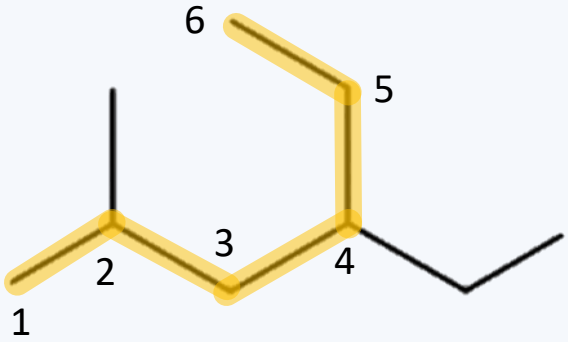
Returned Name: *2,3-dimethylbutane*
Verdict: ✓ ***Match***

☑ longest chain/ring ☑ lowest locants ☑ alphabetical order ☑ punctuation ☑ formula counted

A04 DEVELOPING

C_9H_{20}

1 · NUMBERED PARENT



Parent (6 C) highlighted; carbons numbered from the left end to the top end

2 · NAMING TRAIL

Parent: *longest chain = 6 C → hexane*

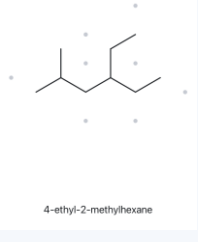
Numbering: *Left End → Top gives {2,4}; Top → Left End gives {3,5}. First point of difference: 2 < 3, so number from the left end*

Substituents: *(A→Z): ethyl on C4; methyl on C2*

3 · MY NAME (WRITTEN FIRST)

4-ethyl-2-methylhexane

4 · THE ALKANE CHECK



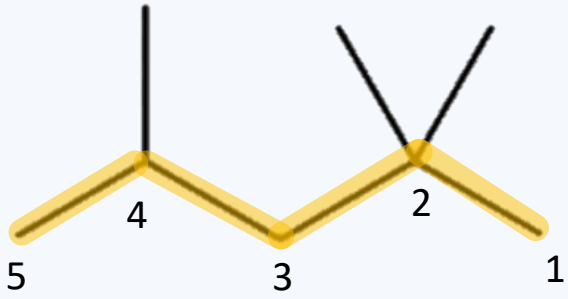
Returned Name: *4-ethyl-2-methylhexane*
Verdict: ✓ ***Match***

☑ longest chain/ring ☑ lowest locants ☑ alphabetical order ☑ punctuation ☑ formula counted

A05 DEVELOPING

C_8H_{18}

1 · NUMBERED PARENT



Parent (5 C) highlighted; carbons numbered from the right to left end

2 · NAMING TRAIL

Parent: *longest chain = 5 C → pentane*

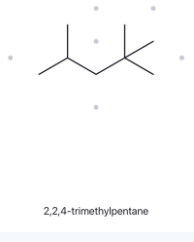
Numbering: *L→R gives {2,4,4}; R→L gives {2,2,4}. First point of difference: 2 < 4 (second locant), so number from right to left*

Substituents: *methyl on C2; methyl on C2; methyl on C4*

3 · MY NAME (WRITTEN FIRST)

2,2,4-trimethylpentane

4 · THE ALKANE CHECK



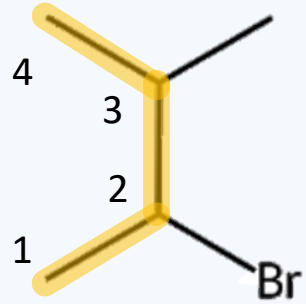
Returned Name: *2,2,4-trimethylpentane*
Verdict: ✓ ***Match***

☑ longest chain/ring ☑ lowest locants ☑ alphabetical order ☑ punctuation ☑ formula counted

A06 DEVELOPING

$C_5H_{11}Br$

1 · NUMBERED PARENT



Parent (4 C) highlighted; carbons numbered from the lower-left end to the upper-left end

2 · NAMING TRAIL

Parent: *longest chain = 4 C → butane*

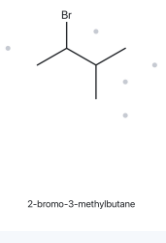
Numbering: *Lower-left → upper-left gives 2,3; upper-left → lower-left gives 2,3. Tie-breaker: bromo (alphabetically first) gets the lower number, so number from lower-left to upper-left*

Substituents: *(A→Z): bromo on C2; methyl on C3*

3 · MY NAME (WRITTEN FIRST)

2-bromo-3-methylbutane

4 · THE ALKANE CHECK



Returned Name: *2-bromo-3-methylbutane*
Verdict: ✓ ***Match***

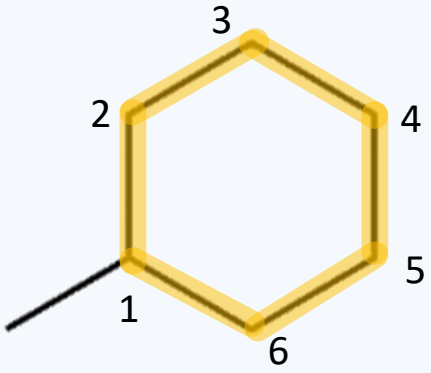
☑ longest chain/ring ☑ lowest locants ☑ alphabetical order ☑ punctuation ☑ formula counted

A07

DEVELOPING

C_7H_{14}

1 · NUMBERED PARENT



Parent (6 C) highlighted; cyclohexane ring numbered starting from the carbon attached to the methyl group (C1), proceeding clockwise

2 · NAMING TRAIL

Parent: *ring = 6 C → cyclohexane*

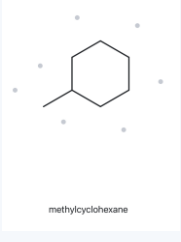
Numbering: *L→R gives 1; R→L gives 1. Only one substituent, so the ring carbon bearing the methyl group is C1, and either direction gives the same locant.*

Substituents: *(A→Z): methyl on C1*

3 · MY NAME (WRITTEN FIRST)

methylcyclohexane

4 · THE ALKANE CHECK



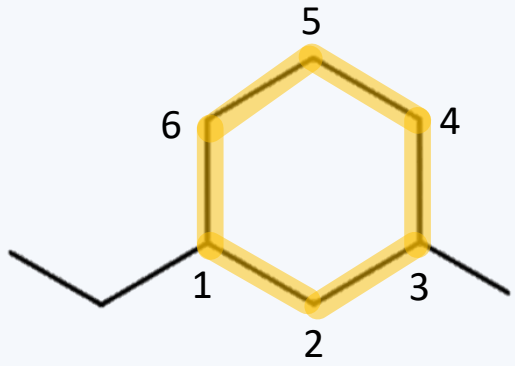
Returned Name: *methylcyclohexane*
Verdict: ✓ ***Match***

☑ longest chain/ring ☑ lowest locants ☑ alphabetical order ☑ punctuation ☑ formula counted

A08 CHALLENGE

C_9H_{18}

1 · NUMBERED PARENT



Parent (6 C) highlighted; cyclohexane ring numbered starting from the carbon attached to the ethyl group (C1), proceeding counterclockwise

2 · NAMING TRAIL

Parent: *ring = 6 C → cyclohexane*

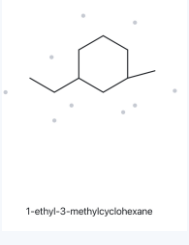
Numbering: *L→R gives 1,3; R→L gives 1,3. The locant sets tie, so ethyl (alphabetically first) takes the lower number: C1 is the ethyl carbon, so number from left to right.*

Substituents: *(A→Z): ethyl on C1; methyl on C3*

3 · MY NAME (WRITTEN FIRST)

1-ethyl-3-methylcyclohexane

4 · THE ALKANE CHECK



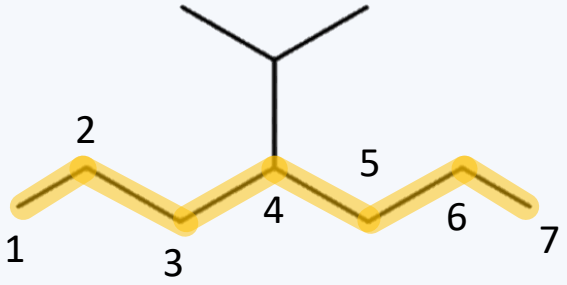
Returned Name: *1-ethyl-3-methylcyclohexane*
Verdict: ✓ ***Match***

☑ longest chain/ring ☑ lowest locants ☑ alphabetical order ☑ punctuation ☑ formula counted

A09 CHALLENGE

$C_{10}H_{22}$

1 · NUMBERED PARENT



Parent (7 C) highlighted; carbons numbered from the left end to the right end

2 · NAMING TRAIL

Parent: *longest chain = 7 C → heptane*

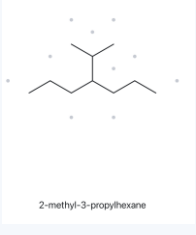
Numbering: *L→R gives 4; R→L gives 4. Both directions give the same locant, so number from left to right.*

Substituents: *(A→Z): isopropyl on C4*

3 · MY NAME (WRITTEN FIRST)

4-isopropylheptane

4 · THE ALKANE CHECK



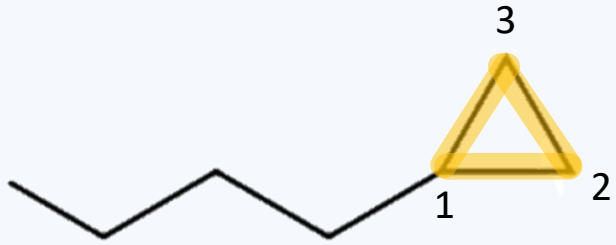
Returned Name: *2-methyl-3-propylhexane*
Verdict: ***X Mismatch***
Correct Name: *4-isopropylheptane (same compound as 4-(1-methylethyl)heptane)*
Rule: *The parent must be the longest continuous chain. The horizontal chain has 7 C, but the tool used a 6-C parent through the isopropyl group.*

☑ longest chain/ring ☑ lowest locants ☑ alphabetical order ☑ punctuation ☑ formula counted

A10 CHALLENGE

C_7H_{14}

1 · NUMBERED PARENT



Parent (ring of 3 C) highlighted; C1 is the ring carbon bearing the butyl group

2 · NAMING TRAIL

Parent: *ring = 3 C → cyclopropane*

Numbering: *Only one substituent, so the ring carbon bearing the butyl group is C1. Either direction around the ring gives the same locant, so no locant is needed in the name*

Substituents: *(A→Z): butyl on C1*

3 · MY NAME (WRITTEN FIRST)

butylcyclopropane

4 · THE ALKANE CHECK



Returned Name: *butylcyclopropane*
Verdict: ✓ ***Match***

☑ longest chain/ring ☑ lowest locants ☑ alphabetical order ☑ punctuation ☑ formula counted

B01 FOUNDATION

2-methylpropane

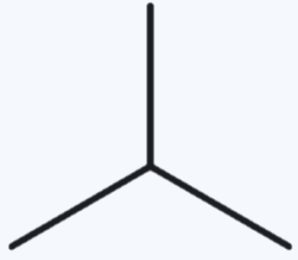
1 · NAME DECODED

Parent: *propane → a 3-carbon chain*

Substituents: *methyl on C2*

Expected Formula: *3 + 1 = 4 C; open chain →* C_4H_{10}

2 · BOND-LINE STRUCTURE (THE ALKANE)



3 · FORMULA CHECK

Counted from the Drawing: C_4H_{10}

Open Chain: $C_nH_{2n+2} = 2(4) + 2 = C_4H_{10}$ ✓

4 · BACK-CHECK

Returned by The Alkane: *2-methylpropane*

✓ ***Same as the Given Name***

☑ parent length ☑ substituent positions ☑ formula agrees ☑ back-check recorded

B02 FOUNDATION

3-ethylpentane

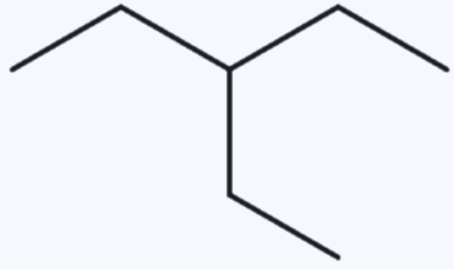
1 · NAME DECODED

Parent: *pentane → a 5-carbon chain*

Substituents: *ethyl on C3*

Expected Formula: *5 + 2 = 7 C; open chain →* C_7H_{16}

2 · BOND-LINE STRUCTURE (THE ALKANE)



3 · FORMULA CHECK

Counted from the Drawing: C_7H_{16}

Open Chain: $C_nH_{2n+2} = 2(7) + 2 = C_7H_{16}$ ✓

4 · BACK-CHECK

Returned by The Alkane: *3-ethylpentane*

✓ ***Same as the Given Name***

☑ parent length ☑ substituent positions ☑ formula agrees ☑ back-check recorded

B03 FOUNDATION

2,2-dimethylbutane

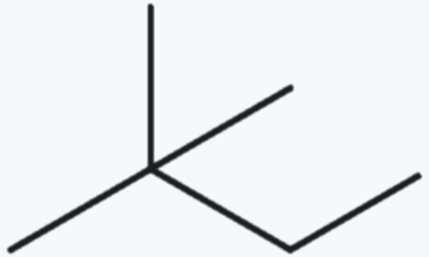
1 · NAME DECODED

Parent: *butane → a 4-carbon chain*

Substituents: *methyl on C2; methyl on C2 (two methyl groups on the same carbon)*

Expected Formula: *4 + 2 = 6 C; open chain →* C_6H_{14}

2 · BOND-LINE STRUCTURE (THE ALKANE)



3 · FORMULA CHECK

Counted from the Drawing: C_6H_{14}

Open Chain: $C_nH_{2n+2} = 2(6) + 2 = C_6H_{14}$ ✓

4 · BACK-CHECK

Returned by The Alkane: *2,2-dimethylbutane*

✓ ***Same as the Given Name***

☑ parent length ☑ substituent positions ☑ formula agrees ☑ back-check recorded

3-ethyl-2-methylpentane

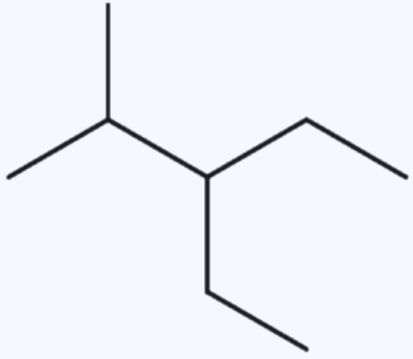
1 · NAME DECODED

Parent: *pentane → a 5-carbon chain*

Substituents: *ethyl on C3; methyl on C2*

Expected Formula: *5 + 2 + 1 = 8 C; open chain →* C_8H_{18}

2 · BOND-LINE STRUCTURE (THE ALKANE)



3 · FORMULA CHECK

Counted from the Drawing: C_8H_{18}

Open Chain: $C_nH_{2n+2} = 2(8) + 2 = C_8H_{18}$ ✓

4 · BACK-CHECK

Returned by The Alkane: *3-ethyl-2-methylpentane*

✓ ***Same as the Given Name***

☑ parent length ☑ substituent positions ☑ formula agrees ☑ back-check recorded

1-chloro-2-methylpropane

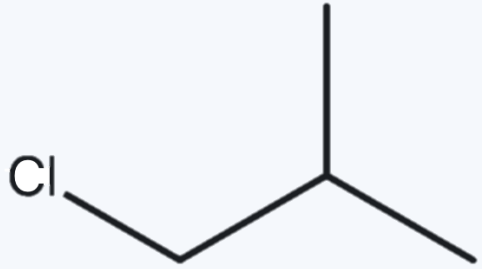
1 · NAME DECODED

Parent: *propane → a 3-carbon chain*

Substituents: *chloro on C1; methyl on C2*

Expected Formula: *3 + 1 = 4 C; open chain → C_4H_{10}; Cl replaces one H → C_4H_9Cl*

2 · BOND-LINE STRUCTURE (THE ALKANE)



3 · FORMULA CHECK

Counted from the Drawing: *C_4H_9Cl*

Open Chain: *C_nH_{2n+2} = 2(4) + 2 = C_4H_{10}; minus one H for Cl = C_4H_9Cl ✓*

4 · BACK-CHECK

Returned by The Alkane: *1-chloro-2-methylpropane*

✓ ***Same as the Given Name***

☑ parent length ☑ substituent positions ☑ formula agrees ☑ back-check recorded

B06 DEVELOPING

1,2-dimethylcyclopentane

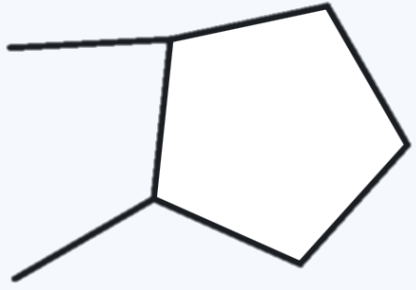
1 · NAME DECODED

Parent: *cyclopentane → a 5-carbon ring*

Substituents: *methyl on C1; methyl on C2*

Expected Formula: *5 + 2 = 7 C; one ring → C_7H_{14}*

2 · BOND-LINE STRUCTURE (THE ALKANE)



3 · FORMULA CHECK

Counted from the Drawing: *C_7H_{14}*

One Ring: *$C_nH_{2n} = C_7H_{14}$* ✓

4 · BACK-CHECK

Returned by The Alkane: *1,2-dimethylcyclopentane*

✓ ***Same as the Given Name***

☑ parent length ☑ substituent positions ☑ formula agrees ☑ back-check recorded

4-ethyl-3,3-dimethylheptane

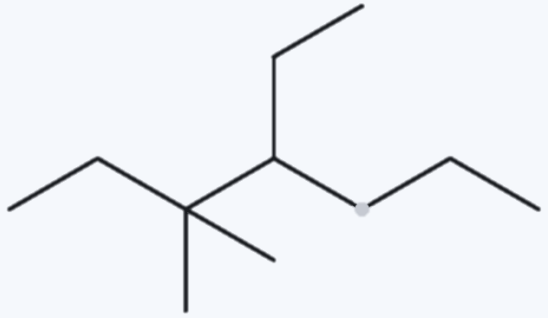
1 · NAME DECODED

Parent: *heptane → a 7-carbon chain*

Substituents: *ethyl on C4; methyl on C3; methyl on C3 (two methyl groups on the same carbon)*

Expected Formula: *7 + 2 + 1 + 1 = 11 C; open chain → $C_{11}H_{24}$*

2 · BOND-LINE STRUCTURE (THE ALKANE)



3 · FORMULA CHECK

Counted from the Drawing: $C_{11}H_{24}$

Open Chain: $C_nH_{2n+2} = 2(11) + 2 = C_{11}H_{24}$ ✓

4 · BACK-CHECK

Returned by The Alkane: *4-ethyl-3,3-dimethylheptane*

✓ ***Same as the Given Name***

☑ parent length ☑ substituent positions ☑ formula agrees ☑ back-check recorded

B08 CHALLENGE

tert-butylcyclohexane

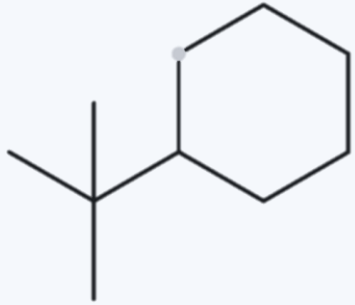
1 · NAME DECODED

Parent: *cyclohexane → a 6-carbon ring*

Substituents: *tert-butyl on C1 (a single substituent on a ring needs no locant). tert-Butyl is a carbon carrying three methyl groups, $(CH_3)_3C–$, 4 carbons in all*

Expected Formula: *6 + 4 = 10 C; one ring → $C_{10}H_{20}$*

2 · BOND-LINE STRUCTURE (THE ALKANE)



3 · FORMULA CHECK

Counted from the Drawing: *$C_{10}H_{20}$*

One Ring: *$C_nH_{2n} = C_{10}H_{20}$ ✓*

4 · BACK-CHECK

Returned by The Alkane: *(1,1-dimethylethyl)cyclohexane*

✓ ***Same as the Given Name*** *(1,1-dimethylethyl) is the systematic form of tert-butyl (Appendix B.3), so the names differ only in how the substituent is written*

☑ parent length ☑ substituent positions ☑ formula agrees ☑ back-check recorded

B09 CHALLENGE

3-ethyl-1,1-dimethylcyclohexane

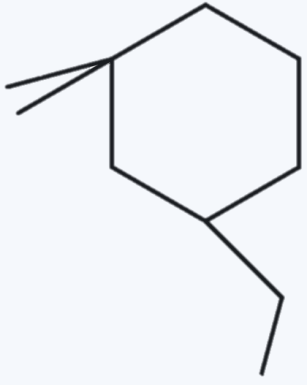
1 · NAME DECODED

Parent: *cyclohexane → a 6-carbon ring*

Substituents: *ethyl on C3; methyl on C1; methyl on C1 (two methyl groups on the same carbon)*

Expected Formula: *6 + 2 + 1 + 1 = 10 C; one ring → $C_nH_{2n} = C_{10}H_{20}$*

2 · BOND-LINE STRUCTURE (THE ALKANE)



3 · FORMULA CHECK

Counted from the Drawing: *$C_{10}H_{20}$*

One Ring: *$C_nH_{2n} = C_{10}H_{20}$* ✓

4 · BACK-CHECK

Returned by The Alkane: *3-ethyl-1,1-dimethylcyclohexane*

✓ ***Same as the Given Name***

☑ parent length ☑ substituent positions ☑ formula agrees ☑ back-check recorded

B10 CHALLENGE

2-ethylbutane

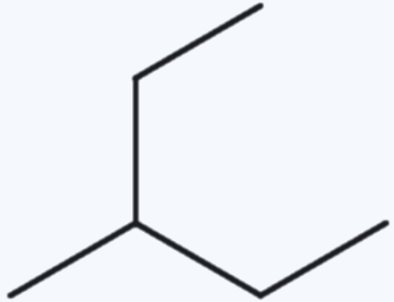
1 · NAME DECODED

Parent: *butane → a 4-carbon chain*

Substituents: *ethyl on C2*

Expected Formula: *4 + 2 = 6 C; open chain →* C_6H_{14}

2 · BOND-LINE STRUCTURE (THE ALKANE)



3 · FORMULA CHECK

Counted from the Drawing: C_6H_{14}

Open Chain: $C_nH_{2n+2} = 2(6) + 2 = C_6H_{14}$ ✓

4 · BACK-CHECK

Returned by The Alkane: *3-methylpentane*

✗ *Differs from the given name*

Correct Name: *3-methylpentane. The given name 2-ethylbutane breaks the longest-chain rule: the ethyl group on C2 extends the chain, so the longest continuous chain in the drawing is 5 carbons (pentane), not 4. The leftover single carbon is a methyl on C3.*

☑ parent length ☑ substituent positions ☑ formula agrees ☑ back-check recorded